TUTORIAL – 8 :-

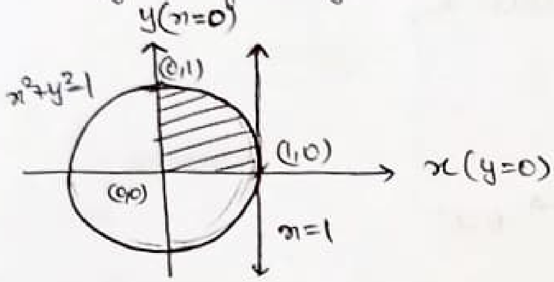
① With the help of a neat diagram mark the region of integration in the following double integrals :

ⓐ $\int_0^1 \int_0^{\sqrt{1-x^2}} f(x,y)\, dy\, dx$

Here x varies from $x=0$ to $x=1$
y varies from $y=0$ to $y=\sqrt{1-x^2}$

$\therefore \quad y^2 = 1-x^2$
$x^2+y^2=1 \longrightarrow$ circle with radius 1

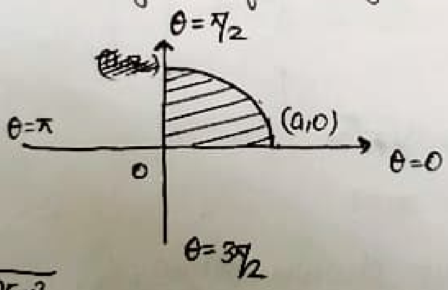
Required region of integration is ,



ⓑ $\int_0^{\pi/2} \int_0^{a} f(r,\theta)\, dr\, d\theta$

Here θ varies from $\theta=0$ to $\theta=\pi/2$
r varies from $r=0$ to $r=a$

Required region of integration is,

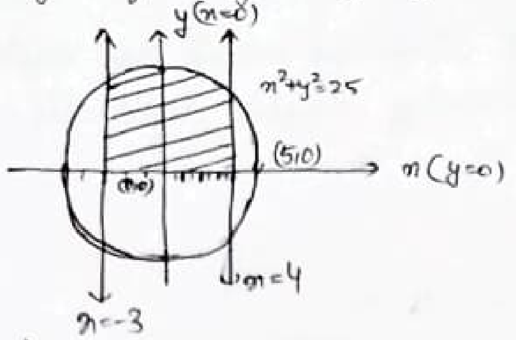


ⓒ $\int_{-3}^{4} \int_0^{\sqrt{25-x^2}} f(x,y)\, dy\, dx$

Here x varies from $x=-3$ to $x=4$
y varies from $y=0$ to $y=\sqrt{25-x^2}$

$\therefore \quad y^2 = 25-x^2$
$x^2+y^2 = 25 \longrightarrow$ circle with radius 5

Required region of integration is,

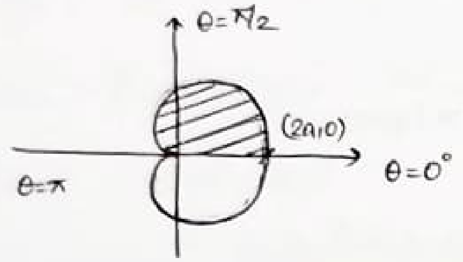


(d) $\int_0^{\pi}\int_0^{a(1+\cos\theta)} f(r,\theta)\,dr\,d\theta$

Here θ varies from $\theta=0$ to $\theta=\pi$

r varies from $r=0$ to $r=a(1+\cos\theta)$

Required region of integration is,



(2) Evaluate the following double integrals:

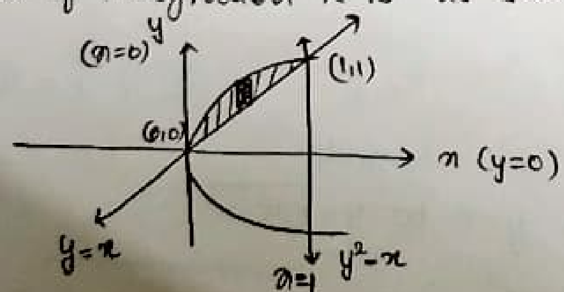
(a) $\int_0^1\int_x^{\sqrt{x}} (x^2+y^2)\,dy\,dx$

Soln given $\int_0^1\int_x^{\sqrt{x}} (x^2+y^2)\,dy\,dx$

Here y varies from $y=x$ to $y=\sqrt{x}$ ($y^2=x$)

x varies from $x=0$ to $x=1$

The region of integration R is as shown below,



$$= \int_0^1 \left(x^2y+\frac{y^3}{3}\right)_x^{\sqrt{x}} dx = \int_0^1 \left(x^{5/2}+\frac{x^{3/2}}{3}-x^3-\frac{x^3}{3}\right)dx$$

$$= \left(\frac{x^{7/2}}{7/2}+\frac{x^{5/2}}{15/2}-\frac{x^4}{4}-\frac{x^4}{12}\right)_0^1 = \frac{2}{7}+\frac{2}{15}-\frac{1}{3} = 3/35$$

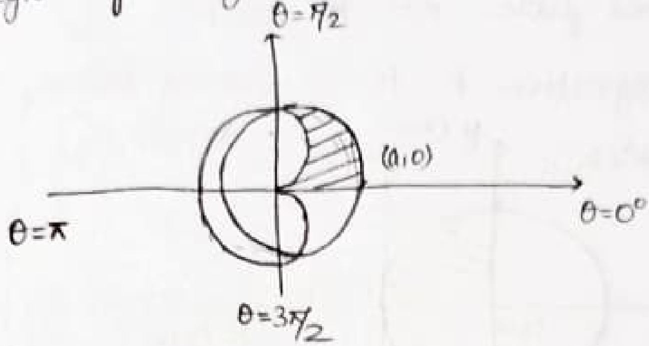
(b) $\int_0^{\pi/2}\int_{a(1-\cos\theta)}^{a} r^2\,dr\,d\theta$

Soln Given $\int_0^{\pi/2}\int_{a(1-\cos\theta)}^{a} r^2\,dr\,d\theta$

Here θ varies from $\theta = 0$ to $\theta = \pi/2$

r varies from $r = a(1-\cos\theta)$ to $r = a$

The region of integration R is as shown below,



$$\Rightarrow \int_0^{\pi/2}\int_{a(1-\cos\theta)}^{a} r^2\,dr\,d\theta$$

$$= \int_0^{\pi/2}\left(\frac{r^3}{3}\right)_{a(1-\cos\theta)}^{a} d\theta$$

$$= \frac{1}{3}\int_0^{\pi/2} a^3\left[1-(1-\cos\theta)^3\right] d\theta$$

$$= \frac{a^3}{3}\int_0^{\pi/2}\left\{1-\left[1-\cos^3\theta-3\cos\theta+3\cos^2\theta\right]\right\} d\theta$$

$$= \frac{a^3}{3}\int_0^{\pi/2}\left(\cos^3\theta+3\cos\theta-3\cos^2\theta\right) d\theta$$

$$= \frac{a^3}{3}\left[\int_0^{\pi/2}\cos^3\theta\,d\theta + 3\int_0^{\pi/2}\cos\theta\,d\theta - 3\int_0^{\pi/2}\cos^2\theta\,d\theta\right]$$

$$= \frac{a^3}{3}\left[\frac{(3-1)(1)}{(3)(3-2)} + 3.\frac{(1)}{(1)} - 3\,\frac{(2-1)}{2\times 1}\times\frac{\pi}{2}\right]$$

$$= \frac{a^3}{3}\left[\frac{2}{3}+3-\frac{3\pi}{4}\right]$$

$$= \frac{a^3}{3}\left[\frac{11}{3}-\frac{3\pi}{4}\right] = a^3\left[\frac{11}{9}-\pi/4\right]$$

ⓒ $\int_0^1 \int_0^{\sqrt{\frac{1-x^2}{2}}} \frac{1}{\sqrt{1-x^2-y^2}}\, dy\, dx$

$2y^2 = 1 + x^2$

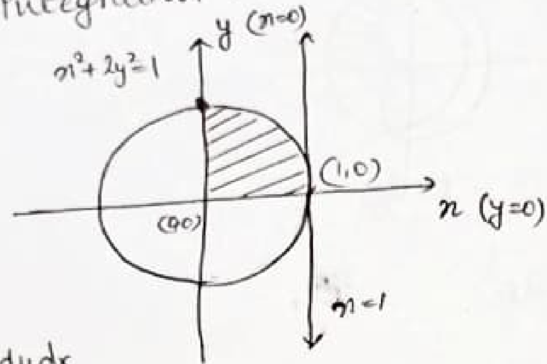
$\left(\frac{x^2}{1} + \frac{y^2}{1/2} = 1\right)$

Soln given $\int_0^1 \int_0^{\sqrt{\frac{1-x^2}{2}}} \frac{1}{\sqrt{1-x^2-y^2}}\, dy\, dx$

Here y varies from $y=0$ to $y = \sqrt{\frac{1-x^2}{2}}$

x varies from $x=0$ to $x=1$

The region of integration R is as shown below,



$$\Rightarrow \int_0^1 \int_0^{\sqrt{\frac{1-x^2}{2}}} \frac{1}{\sqrt{1-x^2-y^2}}\, dy\, dx$$

$$= \int_0^1 \int_0^{\sqrt{\frac{1-x^2}{2}}} \frac{1}{\sqrt{(\sqrt{1-x^2})^2 - y^2}}\, dy\, dx$$

$$= \int_0^1 \left[\sin^{-1}\left(\frac{y}{\sqrt{1-x^2}}\right)\right]_0^{\sqrt{\frac{1-x^2}{2}}} dx$$

$$= \int_0^1 \left[\sin^{-1}\left(\frac{1}{\sqrt{2}}\right) - \sin^{-1}(0)\right] dx$$

$$= \int_0^1 \frac{\pi}{4}\, dx$$

$$= \frac{\pi}{4}(x)_0^1$$

$$= \frac{\pi}{4}$$

③ Evaluate the following triple integrals.

ⓐ $\int_0^1\int_1^2\int_2^3 (x+y+z)\,dx\,dy\,dz$

Solu given $\int_0^1\int_1^2\int_2^3 (x+y+z)\,dx\,dy\,dz$

$$= \int_0^1\int_1^2 \left(\frac{x^2}{2}+xy+xz\right)_2^3 dy\,dz$$

$$= \int_0^1\int_1^2 \left\{\left(\frac{9}{2}+3y+3z\right)-\left(2+2y+2z\right)\right\} dy\,dz$$

$$= \int_0^1\int_1^2 \left(\frac{5}{2}+y+z\right) dy\,dz$$

$$= \int_0^1 \left(\frac{5y}{2}+\frac{y^2}{2}+zy\right)_1^2 dz$$

$$= \int_0^1 \left[(5+2+2z)-\left(\frac{5}{2}+\frac{1}{2}+z\right)\right] dz$$

$$= \int_0^1 (4+z)\,dz$$

$$= \left(4z+\frac{z^2}{2}\right)_0^1$$

$$= 4+\frac{1}{2}-0$$

$$= 9/2$$

ⓑ $\int_{-1}^1\int_0^z\int_{x-z}^{x+z} (x+y+z)\,dy\,dz\,dx$

Solu given $\int_{-1}^1\int_0^z\int_{x-z}^{x+z} (x+y+z)\,dy\,dx\,dz$

$$= \int_{-1}^1\int_0^z \left(xy+\frac{y^2}{2}+zy\right)_{x-z}^{x+z} dx\,dz$$

$$= \int_{-1}^1\int_0^z \left[\left(x^2+xz+\frac{1}{2}(x+z)^2+xz+z^2\right)-\left(x^2-xz+\frac{(x-z)^2}{2}+xz-z^2\right)\right] dx\,dz$$

$$= \int_{-1}^1\int_0^z \left[\cancel{x^2}+2xz+z^2+\frac{\cancel{x^2}}{\cancel{2}}+\frac{\cancel{z^2}}{\cancel{2}}+xz-\cancel{x^2}+z^2+\frac{\cancel{x^2}}{\cancel{2}}-\frac{\cancel{z^2}}{\cancel{2}}+xz\right] dx\,dz$$

$$= \int_{-1}^{1}\int_{0}^{2} (4xz+2z^2)\, dx\,dz$$

$$= \int_{-1}^{1} \left(2x^2z+2z^2x\right)_0^z dz$$

$$= \int_{-1}^{1} \left(2z^3+2z^3\right) dz$$

$$= 4\int_{-1}^{1} z^3\, dz$$

$$= 4\left(\frac{z^4}{4}\right)_{-1}^{1}$$

$$= 1-1$$

$$= 0$$

ⓒ $$\int_0^1\int_0^{\sqrt{1-x^2}}\int_0^{\sqrt{1-x^2-y^2}} \frac{1}{\sqrt{1-x^2-y^2-z^2}}\, dz\,dy\,dx$$

Soln given $$\int_0^1\int_0^{\sqrt{1-x^2}}\int_0^{\sqrt{1-x^2-y^2}} \frac{1}{\sqrt{1-x^2-y^2-z^2}}\, dz\,dy\,dx$$

$$= \int_0^1\int_0^{\sqrt{1-x^2}}\int_0^{\sqrt{1-x^2-y^2}} \frac{1}{\sqrt{\left(\sqrt{1-x^2-y^2}\right)^2-z^2}}\, dz\,dy\,dx$$

$$= \int_0^1\int_0^{\sqrt{1-x^2}} \left[\sin^{-1}\left(\frac{z}{\sqrt{1-x^2-y^2}}\right)\right]_0^{\sqrt{1-x^2-y^2}} dy\,dx$$

$$= \int_0^1\int_0^{\sqrt{1-x^2}} \frac{\pi}{2}\, dy\,dx$$

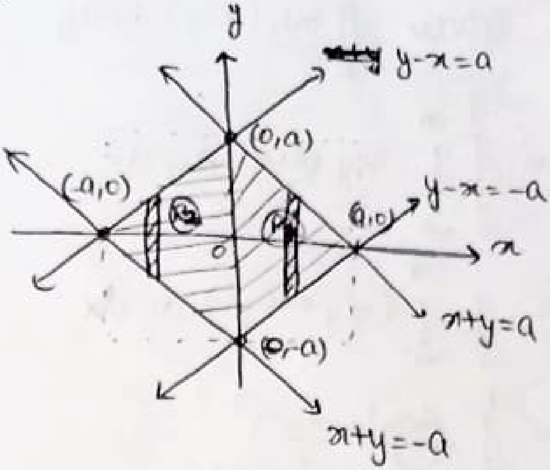
$$= \int_0^1 \left[\frac{\pi y}{2}\right]_0^{\sqrt{1-x^2}} dx$$

$$= \frac{\pi}{2}\int_0^1 \sqrt{1-x^2}\, dx$$

$$= \frac{\pi}{2}\left[\frac{x}{2}\sqrt{1-x^2}+\frac{1}{2}\sin^{-1}(x)\right]_0^1 = \frac{\pi}{2}\left[\frac{1}{2}(0)+\frac{1}{2}\sin^{-1}(1)\right] = \frac{\pi}{2}\times\frac{\pi}{4} = \frac{\pi^2}{8}$$

TUTORIAL 9 :

① Evaluate $\iint_R e^{x+y}\,dx\,dy$, where R is the region bounded by the lines $x+y=\pm a$ and $y-x=\pm a$.



Soln Given $\iint_R e^{x+y}\,dx\,dy$

$$R_1=\int_0^a\int_{x-a}^{a-x} e^{x+y}\,dy\,dx$$

$$=\int_0^a e^x\left(e^y\right)_{x-a}^{a-x}dx$$

$$=\int_0^a e^x\left(e^{a-x}-e^{-(a-x)}\right)dx$$

$$=\int_0^a e^a-e^{2x-a}\,dx$$

$$=\left(xe^a-\frac{e^{2x-a}}{2}\right)_0^a \quad = ae^a-\frac{e^a}{2}+\frac{e^{-a}}{2}$$

$$R_2=\int_{-a}^0\int_{-(x+a)}^{x+a} e^{x+y}\,dy\,dx$$

$$=\int_{-a}^0 e^x\left(e^y\right)_{-(x+a)}^{x+a}dx$$

$$=\int_{-a}^0 e^x\left[e^{x+a}-e^{-(x+a)}\right]dx$$

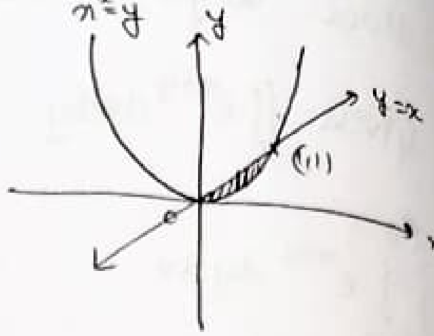
$$=\int_{-a}^0\left(e^{2x+a}-e^{-a}\right)dx$$

$$=\left[\frac{e^{2x+a}}{2}-xe^{-a}\right]_{-a}^0$$

$$=\frac{e^a}{2}-\frac{e^{-a}}{2}-ae^{-a}$$

$$\therefore \iint_R e^{x+y}\,dx\,dy = R_1+R_2 = \underline{\underline{a\left(e^a-e^{-a}\right)}}$$

(2) Evaluate $\iint_R xy(x+y)\,dx\,dy$ where R is the region bounded between $y=x^2$ and $y=x$.



Solu: Given $\iint_R xy(x+y)\,dx\,dy$

$$= \int_0^1\int_{x^2}^{x} xy(x+y)\,dy\,dx$$

$$= \int_0^1\int_{x^2}^{x} (x^2y+xy^2)\,dy\,dx$$

Here x varies from zero to one
y varies from $y=x^2$ to $y=x$

$$= \int_0^1 \left(\frac{x^2y^2}{2}+\frac{xy^3}{3}\right)_{x^2}^{x}$$

$$= \int_0^1 \frac{x^4}{2}+\frac{x^4}{3}-\frac{x^6}{2}-\frac{x^7}{3}\,dx$$

$$= \int_0^1 \left(\frac{5}{6}x^4-\frac{1}{2}x^6-\frac{1}{3}x^7\right)dx$$

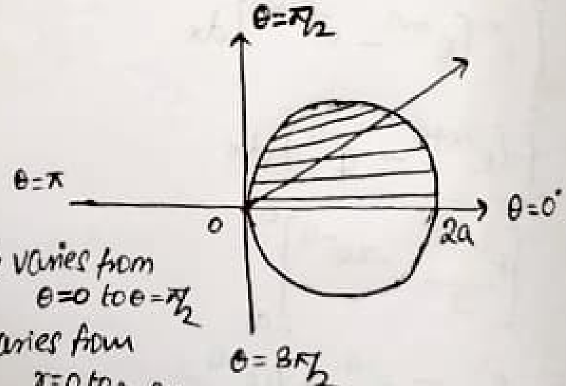
$\frac{1}{8}-\frac{1}{14}$
$=\frac{6}{8\times 14}$

$$= \left(\frac{5}{6}\left(\frac{x^5}{5}\right)-\frac{1}{2}\left(\frac{x^7}{7}\right)-\frac{1}{3}\left(\frac{x^8}{8}\right)\right)_0^1$$

$$= \frac{1}{6}-\frac{1}{14}-\frac{1}{24}$$

$$= 3/56$$

(3) Evaluate $\iint_R r^2\sin\theta\,dr\,d\theta$ where R is the region bounded by semi-circle $r=2a\cos\theta$ above the initial line.



Solu: Given $\iint_R r^2\sin\theta\,dr\,d\theta$

$$= \int_0^{\pi/2}\int_0^{2a\cos\theta} r^2\sin\theta\,dr\,d\theta$$

Here θ varies from
θ=0 to θ=π/2
r varies from
r=0 to r=2acosθ

$$= \int_0^{\pi/2} \sin\theta\left[\frac{r^3}{3}\right]_0^{2a\cos\theta} d\theta$$

$$= \int_0^{\pi/2} \frac{\sin\theta}{3}\,8a^3\cos^3\theta\,d\theta = \frac{8a^3}{3}\int_0^{\pi/2}\sin\theta\cos^3\theta\,d\theta$$

$$= \frac{8a^3}{3}\left(\frac{(1\times(3-1))}{4\times2\times1}\right) = \frac{8a^3}{3}\,\frac{2}{4\times2\times1} = \frac{2a^3}{3}$$

5. By changing the order of integration, evaluate the following double integrals :

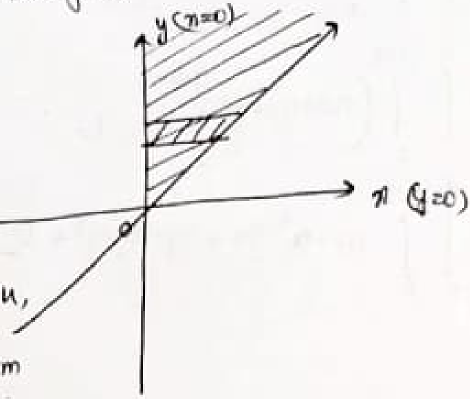
(a) $\int_0^\infty \int_x^\infty \frac{e^{-y}}{y}\, dy\, dx$

Soln Given $\int_0^\infty \int_x^\infty \frac{e^{-y}}{y}\, dy\, dx$

Here x varies from $x=0$ to $x=\infty$

y varies from $y=x$ to $y=\infty$

The region of integration R is given as below,



By changing the order of integration,

Here x varies from $x=0$ to $x=y$

y varies from $y=0$ to $y=\infty$

$$\therefore \int_0^\infty \int_x^\infty \frac{e^{-y}}{y}\, dy\, dx = \int_0^\infty \int_0^y \frac{e^{-y}}{y}\, dx\, dy$$

$$= \int_0^\infty \frac{e^{-y}}{y} (x)_0^y\, dy$$

$$= \int_0^\infty e^{-y}\, dy$$

$$= (-e^{-y})_0^\infty$$

$$= (-e^{-\infty} + e^0)$$

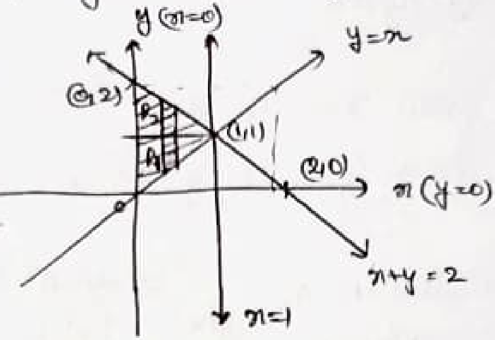
$$= 1$$

(b) $\int_0^1 \int_x^{2-x} xy \, dy\,dx$

Soln: Given $\int_0^1 \int_x^{2-x} xy \, dy\,dx$

Here x varies from $x=0$ to $x=1$
y varies from $y=x$ to $y=2-x$

The region of integration R is given as below,



By changing the order of integration,

Here, in region 1,
x varies from $x=0$ to $x=y$
y varies from $y=0$ to $y=1$

In region 2,
x varies from $x=0$ to $x=2-y$
y varies from $y=1$ to $y=2$

$R_1 = \int_0^1 \int_0^y xy \, dx\,dy$

$= \int_0^1 \left(\frac{x^2 y}{2}\right)_0^y dy$

$= \int_0^1 \frac{y^3}{2} dy$

$= \left(\frac{y^4}{8}\right)_0^1$

$= 1/8$

$R_2 = \int_1^2 \int_0^{2-y} xy \, dx\,dy$

$= \int_1^2 \left[\frac{x^2 y}{2}\right]_0^{2-y} dy$

$= \int_1^2 \frac{y}{2} (2-y)^2 dy$

$= \int_1^2 \frac{y}{2} (4+y^2-4y) \, dy$

$= \int_1^2 \left(2y + \frac{y^3}{2} - 2y^2\right) dy$

$= \left(y^2 + \frac{y^4}{8} - \frac{2y^3}{3}\right)_1^2$

$= \left(4+\frac{16}{8}-\frac{16}{3}\right) - \left(1+\frac{1}{8}-\frac{2}{3}\right)$

$= 3 + \frac{15}{8} - \frac{14}{3}$

$\therefore \int_0^1 \int_x^{2-x} xy \, dy\,dx = R_1 + R_2$

$= \frac{1}{8} + 3 + \frac{15}{8} - \frac{14}{3}$

$= 5 - \frac{14}{3} = 1/3$

TUTORIAL - 10 :

① Write the limits of integration with respect to r, θ while evaluating the following integrals :

ⓐ $\int_0^a \int_0^{\sqrt{a^2-x^2}} f(x,y)\, dy\, dx$

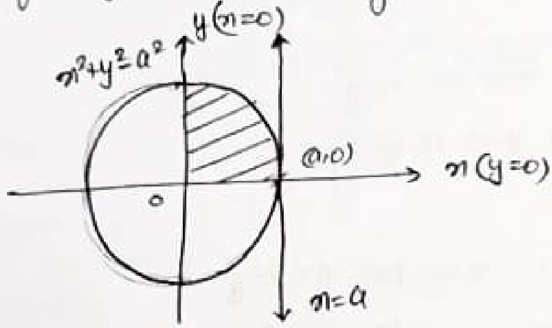
Soln: Given $\int_0^a \int_0^{\sqrt{a^2-x^2}} f(x,y)\, dy\, dx$

Here x varies from $x=0$ to $x=a$

y varies from $y=0$ to $y=\sqrt{a^2-x^2}$

$(x^2+y^2=a^2)$

The region of integration R as given below is



Let $x = r\cos\theta$, $y = r\sin\theta$

$\therefore x^2+y^2 = r^2(\cos^2\theta+\sin^2\theta)$

$x^2+y^2 = r^2$

Also $dx\,dy = r\,dr\,d\theta$

$\therefore$ θ varies from $\theta=0$ to $\theta=\pi/2$

r varies from $r=0$ to $r=a$

$\therefore$ In polar form,

$$\boxed{\int_0^{\pi/2}\int_0^a f(r\cos\theta, r\sin\theta)\, r\, dr\, d\theta}$$

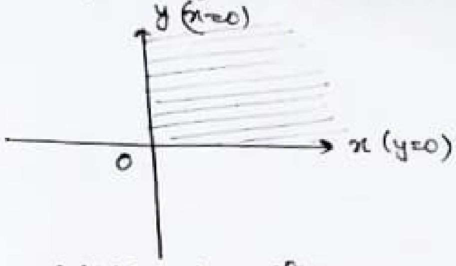
ⓑ $\int_0^\infty \int_0^\infty f(x,y)\, dy\, dx$

Soln given $\int_0^\infty \int_0^\infty f(x,y)\, dy\, dx$

Here x varies from $x=0$ to $x=\infty$

y varies from $y=0$ to $y=\infty$

The region of integration R as given below is,



Let $x = r\cos\theta$, $y = r\sin\theta$

$\therefore$ $x^2+y^2 = r^2(\sin^2\theta+\cos^2\theta)$

$x^2+y^2 = r^2$

Also $dx\,dy = r\,dr\,d\theta$

$\therefore$ θ varies from $\theta=0$ to $\theta=\pi/2$

r varies from $r=0$ to $r=\infty$

$\therefore$ in polar form

$$\boxed{\int_0^{\pi/2}\int_0^{\infty} f(r\cos\theta, r\sin\theta)\, r\,dr\,d\theta}$$

(2) Using the transformation $x+y=u$, $x-y=v$, evaluate the double integral $\iint_R (x^2+y^2)\,dx\,dy$ where R is the region bounded by $u=0$, $u=2$, $v=0$, $v=2$

Soln: Given $\iint_R (x^2+y^2)\,dx\,dy$

$x+y=u$, $x-y=v$

Adding both, we get,

$2x = u+v$

$x = \frac{u+v}{2}$

Subtracting both, we get,

$2y = u-v$

$y = \frac{u-v}{2}$

$\therefore$ $x^2+y^2 = \frac{1}{4}\left[(u+v)^2+(u-v)^2\right] = \frac{1}{2}(u^2+v^2)$,

$$\iint_R (x^2+y^2)\,dx\,dy = \int_{v=0}^{2}\int_{u=0}^{2} \frac{(u^2+v^2)}{2}\,du\,dv$$

$$= \frac{1}{2}\int_0^2 \left(\frac{u^3}{3}+v^2u\right)_0^2 dv$$

$$= \frac{1}{2}\int_0^2 \left(\frac{8}{3}+2v^2\right) dv$$

$$= \frac{1}{2}\left(\frac{8v}{3}+\frac{2v^3}{3}\right)_0^2$$

$$= \frac{1}{2}\left(\frac{16}{3}+\frac{16}{3}\right)$$

$$= \frac{16}{3}$$

③ Evaluate the following integrals by changing to polar coords.

(a) $\int_0^a \int_0^{\sqrt{a^2-x^2}} y\sqrt{x^2+y^2}\,dy\,dx$

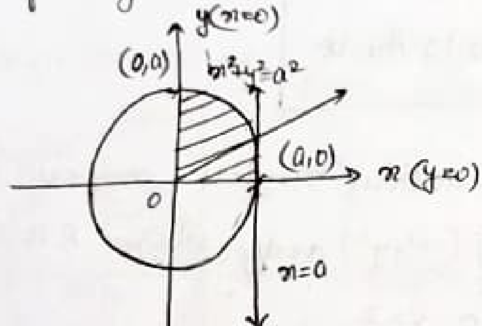
Soln Given $\int_0^a \int_0^{\sqrt{a^2-x^2}} y\sqrt{x^2+y^2}\,dy\,dx$

Here x varies from $x=0$ to $x=a$

y varies from $y=0$ to $y=\sqrt{a^2-x^2}$

$x^2+y^2=a^2$

The region of integration R is as shown below,



Let $x=r\cos\theta,\ y=r\sin\theta$

Also $x^2+y^2=r^2$ and $dx\,dy = r\,dr\,d\theta$

Here θ varies from $\theta=0$ to $\theta=\pi/2$

r varies from $r=0$ to $r=a$

$$\therefore \int_0^{\pi/2}\int_0^a r\sin\theta.(r)\cdot r\,dr\,d\theta$$

$$= \int_0^{\pi/2}\int_0^a r^3\sin\theta\,dr\,d\theta$$

$$= \int_0^{\pi/2} \frac{a^4}{4}\sin\theta\,d\theta$$

$$= \frac{a^4}{4}$$

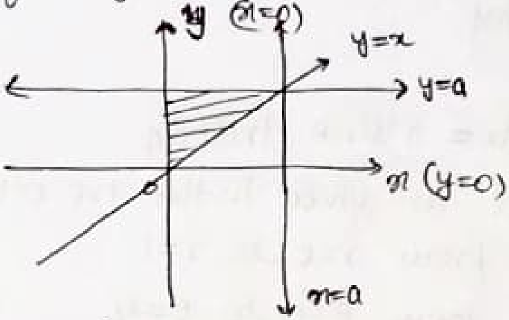
(b) $\int_0^a \int_x^a \frac{x}{x^2+y^2}\, dy\, dx$

Soln Given $\int_0^a \int_x^a \frac{x}{x^2+y^2}\, dy\, dx$

Here x varies from $x=0$ to $x=a$

y varies from $y=x$ to $y=a$

The region of integration R is as shown below,



Let $x = r\cos\theta$, $y = r\sin\theta$

Also $x^2+y^2 = r^2(\sin^2\theta + \cos^2\theta)$

$x^2+y^2 = r^2$ and $dx\,dy = r\,dr\,d\theta$

Here θ varies from $\theta = \pi/4$ to $\theta = \pi/2$

r varies from $r=0$ to $r = \frac{a}{\sin\theta}$

$$\therefore \int_{\pi/4}^{\pi/2} \int_0^{a/\sin\theta} \frac{r\cos\theta}{r^2} \cdot r\, dr\, d\theta$$

$$= \int_{\pi/4}^{\pi/2} \int_0^{a/\sin\theta} \cos\theta\, dr\, d\theta$$

$$= \int_{\pi/4}^{\pi/2} a\cot\theta\, d\theta$$

$$= a\left(\log_e \sin\theta\right)_{\pi/4}^{\pi/2}$$

$$= a\log_e \sqrt{2}$$

$$\frac{a}{2}\ln 2$$

(4) Evaluate $\displaystyle\int_0^1\int_0^{\sqrt{1-x^2}}\int_0^{\sqrt{1-x^2-y^2}} \frac{1}{\sqrt{1-x^2-y^2-z^2}}\,dx\,dy\,dz$, by changing into spherical polar coordinates.

Soln given $\displaystyle\int_0^1\int_0^{\sqrt{1-x^2}}\int_0^{\sqrt{1-x^2-y^2}} \frac{1}{\sqrt{1-x^2-y^2-z^2}}\,dz\,dy\,dx$

Put $x = r\sin\theta\cos\phi$

$y = r\sin\theta\sin\phi$

$z = r\cos\theta$

so that $dz\,dy\,dx = r^2\sin\theta\,dr\,d\theta\,d\phi$

Since the limits are given in the +ve octant,

r varies from $r=0$ to $r=1$

θ varies from $\theta=0$ to $\theta=\pi/2$

ϕ varies from $\phi=0$ to $\phi=\pi/2$

$$\therefore \int_0^1\int_0^{\sqrt{1-x^2}}\int_0^{\sqrt{1-x^2-y^2}} \frac{1}{\sqrt{1-x^2-y^2-z^2}}\,dz\,dy\,dx = \int_0^{\pi/2}\int_0^{\pi/2}\int_0^1 \frac{1}{\sqrt{1-r^2}}\,r^2\sin\theta\,dr\,d\theta\,d\phi$$

$$= \int_0^{\pi/2}\int_0^{\pi/2}\int_0^1 \frac{1-(1-r^2)}{\sqrt{1-r^2}}\sin\theta\,dr\,d\theta\,d\phi$$

$$= \int_0^{\pi/2}\int_0^{\pi/2}\int_0^1 \frac{1}{\sqrt{1-r^2}} - \sqrt{1-r^2}\,(\sin\theta)\,dr\,d\theta\,d\phi$$

$$= \int_0^{\pi/2}\int_0^{\pi/2}\left[\sin^{-1}(r) - \left\{\frac{r}{2}\sqrt{1-r^2} + \frac{1}{2}\sin^{-1}(r)\right\}\right]_0^1 \sin\theta\,d\theta\,d\phi$$

$$= \int_0^{\pi/2}\int_0^{\pi/2}\left[\pi/2 - \pi/4\right]\sin\theta\,d\theta\,d\phi$$

$$= \frac{\pi}{4}\int_0^{\pi/2}(-\cos\theta)_0^{\pi/2}\,d\phi$$

$$= \frac{\pi}{4}\int_0^{\pi/2} 1\,d\phi$$

$$= \pi^2/8$$